EQUIPMENT CONDITION MONITORING

Algorithm

Step 1: Begin.

Step 2: Allow the user to enter the equipment ID, temperature in °C and the vibration in mm/s. Store each value in a separate variable so that they can be used later in the checks.

Step 3: Check the temperature against the monitoring limits. If the temperature is below 80 °C, display the NORMAL message: “Equipment operating within defined monitoring limits.”

Step 4: If the temperature is not below 80 °C, check whether it is less than or equal to 100 °C. If it is, display the WARNING message that the equipment temperature is approaching the predefined threshold.

Step 5: If the temperature is above 100 °C, display the CRITICAL message that the equipment temperature exceeds the predefined threshold, and then display “Inspection required”.

Step 6: Check the vibration against the monitoring limits. If the vibration is below 5 mm/s, display the NORMAL message: “Equipment operating within defined monitoring limits.”

Step 7: If the vibration is not below 5 mm/s, check whether it is less than or equal to 8 mm/s. If it is, display the WARNING message that the equipment vibration is approaching the predefined threshold.

Step 8: If the vibration is above 8 mm/s, display the CRITICAL message that the equipment vibration exceeds the predefined threshold, and then display “Inspection required”.

Step 9: Display the equipment ID so that the user knows which equipment the results belong to, and below it, display all the relevant messages that align with that equipment.

Step 10: End.

Pseudocode

```
FUNCTION CheckEquipmentCondition ():
    INPUT equipment_id
    INPUT temperature
    INPUT vibration


FUNCTION CheckEquipmentCondition (temperature, vibration):
    IF temperature < 80:
        display "NORMAL: Equipment operating within defined monitoring limits."
    ELSE:
        display "WARNING: Equipment temperature approaching predefined threshold."
        display "Inspection required"


    IF vibration < 5:
        display "NORMAL: Equipment operating within defined monitoring limits."
    ELSE IF vibration <= 8:
        display "WARNING: Equipment temperature approaching predefined threshold."
        display "Inspection required"
    ELSE:
        display "CRITICAL: Equipment vibration exceeds predefined threshold."
        display "Inspection required"
END FUNCTION
```